⭐ TOPIC 00: ORIENTATION & PHILOSOPHY

1 PURPOSE

Zero Fluff. Pure FAANG Signal.

Engineered for Senior & Staff SWE offers at Google, Meta, Amazon, Uber, Datadog & Airbnb.

Outcome: Instant pattern recognition + zero-bug Java templates.

2 TARGET AUDIENCE

🌱 **Beginners:** Intuitive mental models without heavy math proofs.

💼 **FAANG Candidates:** Master NeetCode 150 & LC Hard patterns.

⚡ **Staff Engineers:** 20-min pre-interview template cram sheet.

3 WHY THIS HANDBOOK IS DIFFERENT

University / Textbooks	This FAANG Handbook
Isolated proofs & history	**Pattern Evolution & DNA**
Generic pseudocode	**Production-Grade Java 17+**
Memorizing 1000 problems	**29 Core Pattern DNA Cards**
Ignores interviews	**Verbatim Interview Scripts**

💡 AHA MOMENT

Pattern DNA > Problem Count: Solving 1,000 random problems is a trap. Understanding 29 Pattern DNA cards solves thousands automatically!

🏆 GOLDEN RULE

CLARIFY → THINK → CHOOSE DS → OPTIMIZE → CODE → EXPLAIN. Never start coding without stating your pattern!

☕ JAVA ONLY

100% Java Collections API (ArrayDeque, PriorityQueue, HashMap). No generic pseudocode excuses.

⚡ SPEED DRILL

Write every template from memory in under 3 minutes. Zero syntax hesitation under pressure.

📌 **LEARNING PHILOSOPHY:** "Build intuition first, formalize into a template second, dry run on edge cases third." Turn the page to see the complete 20-Topic Dependency Graph!

🗺️ TOPIC 00: SEQUENCED ROADMAP & DEPENDENCY TREE

PHASE 1: LINEAR & SEARCH

- **00** Orientation & Strategy — Start
- ↓
- **01** Foundations (Big-O, Math, Bits) — Base
- ↓
- **02** Arrays & Prefix Sum — Core
- ↓
- **03** Strings & KMP / Hash — Core
- ↓
- **04** Hashing & Frequency — Core
- ↓
- **05** Two Pointers — Pattern
- ↓
- **06** Sliding Window — Pattern
- ↓
- **07** Binary Search & Bounds — Search

PHASE 2: STRUCTURES & TREES

- **08** Linked Lists (Fast/Slow) — Struct
- ↓
- **09** Stack & Monotonic Stack — Mono
- ↓
- **10** Queue & Monotonic Deque — Mono
- ↓
- **11** Heap / PriorityQueue — Top-K
- ↓
- **12** Trees (DFS / BFS / BST) — Heavy
- ↓
- **13** Trie (Prefix Trees) — String

PHASE 3: GRAPHS, DP & META

- **14** Graphs (DFS, BFS, Topo, UF) — Heavy
- ↓
- **15** Recursion & Backtracking — Search
- ↓
- **16** Dynamic Programming — Heavy
- ↓
- **17** Greedy Algorithms — Opt
- ↓
- **18** Staff Data Structures (LRU/LFU) — Staff
- ↓
- **19** Interview Mastery & Meta — Final

⚠️ **DEPENDENCY RULE:** Never jump to Graphs (14) or DP (16) without mastering Trees (12) and Recursion (15). DP is simply Recursion + State Caching!

🏗️ HOW EVERY CHAPTER WORKS (THE 13-BLOCK FLOW)

1 RECOGNITION

Trigger keywords in problem statements mapping instantly to this pattern.

2 MENTAL MODEL

Intuitive visual/physical representation of how the algorithm operates.

3 VISUALIZATION

Step-by-step state diagrams, ASCII trees, and pointer movement illustrations.

4 CORE THEORY

Mathematical invariants and precise structural mechanics behind the pattern.

5 JAVA TEMPLATE

Production-grade Java code skeleton (cross-referenced to Appendix A).

6 LINE-BY-LINE

Detailed explanation of every variable, loop condition, and base case.

7 DRY RUN

Line-by-line variable state trace table on concrete representative inputs.

8 COMPLEXITY

Time & Space complexity analysis with mathematical justification.

9 EDGE CASES

Tricky boundary conditions (null, empty arrays, duplicates, single elements).

10 COMMON MISTAKES

Top 3 failure points candidate make in live FAANG coding interviews.

11 INTERVIEW QUESTIONS

Senior-level follow-up questions asked by Google, Meta & Uber interviewers.

12 LEETCODE SET

Curated progression of representative problems (Easy → Medium → Hard).

13 REVISION SUMMARY SHEET

1-Page condensed cheat card for 20-minute rapid pre-interview revision.

🎯 **CONSISTENCY GUARANTEE:** Every single chapter in Book 1 & Book 2 follows this exact 13-block sequence so you never waste cognitive energy figuring out document formatting!

TOPIC 00: STUDY GUIDE & EXECUTION PLANS

30-DAY FAST TRACK

Experienced Revisers

- **Days 1–7:** Topics 01–07
- **Days 8–18:** Topics 08–14
- **Days 19–25:** Topics 15–17
- **Days 26–30:** Template Drills

60-DAY STANDARD

Recommended FAANG

- **Days 1–15:** Linear & Search
- **Days 16–35:** Trees & Graphs
- **Days 36–50:** Complete DP
- **Days 51–60:** Book 2 Mocks

90-DAY MASTERY

Beginners / Staff

- **Month 1:** Foundations
- **Month 2:** Non-Linear Data
- **Month 3:** Advanced DP & Mocks

INTERVIEW WEEK & 3-HOUR CRAM STRATEGY

Timeframe	Action Plan	Target Resource
7 Days Before	Review all 29 Pattern DNA cards in Book 2 Section 2	Book 2 Section 2
3 Days Before	Write all 24 Master Java Templates from memory	Book 1 Appendix A
1 Day Before	Read "If You See... Think..." Tables & Common Mistakes	Part 13 (Interview Meta)
3 Hours Before	Run 3-Hour Cram Sequence (Topic 00 Page 3)	Topic 00 Page 3 Table

READINESS CHECKLIST: You have completed Topic 00! You understand the 20-Topic Dependency Graph, the 13-Block Chapter Architecture, and your personal Study Schedule. Turn the page to **Topic 01: Foundations (Big-O, Math & Bit Manipulation)**!